

Advanced Time Series Analysis and Forecasting

Seminar 8: Advanced volatility modelling: realised measures, HAR and multivariate GARCH



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BV**: Bipower Variation
- **CDS**: Credit Default Swap
- **CLT**: Central Limit Theorem
- **DCC**: Dynamic Conditional Correlation
- **DCC-NL**: DCC with Nonlinear shrinkage of the target (Engle–Ledoit–Wolf 2019)
- **DM**: Diebold–Mariano (test)
- **EODHD**: EOD Historical Data (market-data provider)
- **ETF**: Exchange-Traded Fund
- **FTX**: FTX (cryptocurrency exchange, bankrupt in November 2022)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-MIDAS**: GARCH with a MIxed DAta Sampling long-run component (Engle–Ghysels–Sohn 2013)

Acronyms used in this material (2/2)

- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HARQ**: HAR with realised Quarticity
- **IQ**: Integrated Quarticity
- **ML**: Machine Learning
- **MSE**: Mean Squared Error
- **OLS**: Ordinary Least Squares
- **OPG**: Outer Product of the Gradient (scores)
- **QLIKE**: Quasi-LIKElihood loss
- **QML**: Quasi-Maximum Likelihood
- **QMLE**: Quasi-Maximum Likelihood Estimator
- **RK**: Realised Kernel
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SCA**: Seasonally and Calendar Adjusted
- **TQ**: Tripower Quarticity
- **TSA**: Time Series Analysis
- **TSRV**: Two-Scales Realised Variance
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VR**: Variance Ratio

Today's question and seminar structure

- The question of the seminar
  - **how much do we really know about a volatility estimate, and when does a better measurement give a better forecast?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	robust standard error of a QML variance; a sandwich by hand	Solved, Proposed	A1
A3, A4	noise bias and optimal sampling; two scales and Parzen weights	Solved, Proposed	A3
A5, A6	RV, BV and the jump test on a toy day; the smallest detectable jump	Solved, Proposed	A5
A7, A8	HAR as a restricted AR(22); attenuation and HARQ	Solved, Proposed	A7
A9, A10	robust and non-robust losses; one DCC and one cDCC step	Solved, Proposed	A9
B1, B2	jumps in Ether; HAR for the DAX	Solved	–
B3	HARQ for Ether out of sample	Proposed	B2, A8
B4, B5	QML for EUR/RON; GARCH-MIDAS for the BET	Solved, Proposed	B4
C1, C2	Bitcoin before and after the spot ETFs; what is wrong in an AI answer?	Proposed	B1–B5

- **[Solved]**: the full solution is in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it following the model, and the solution is discussed in class
- Part A: calculations and derivations on paper; Part B: real data, solved in the notebook, each ending with an interpretation question; Part C: open questions, including the critique of an AI answer

Data used

Series	Source	Frequency	Period
Ether, Bitcoin: realised measures	Binance, one-minute prices	daily	2018–2026
DAX: 5-minute RV and BV	Heber, Lunde, Shephard and Sheppard (2009)	daily	2000–2022
EUR/RON reference rate; BET	BNR; EODHD	daily	2003–2026
Romanian industrial production (SCA)	Eurostat sts_inpr_m	monthly	2000–2026
S&P 500	EODHD	daily	2021–2026

- Functions in the notebook
  - `fit_garch`, `sandwich`, `bipower`, `tripower`, `ht_stat`, `har_X`, `har_forecasts`, `qlike`, `dm_test`, `fit_garch_midas`
- No account or key is needed

Prerequisites for Today (1/23): GARCH and quasi-maximum likelihood

- GARCH(1,1) (TSA, Chapter 5): the return shock is a volatility times a standardised shock

$$\varepsilon_t = \sigma_t \eta_t, \qquad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

  - ε_t: demeaned return of day t; σ_t^2: its conditional variance given the past; $\omega > 0$, $\alpha, \beta \geq 0$: parameters, $\theta = (\omega, \alpha, \beta)$
  - η_t: i.i.d. shock with mean 0 and variance 1; $\kappa_\eta = \mathbb{E}\eta_t^4$: its kurtosis (3 for the Normal distribution)
- QML: maximise the Gaussian log-likelihood even if η_t is not Gaussian

$$\hat{\theta} = \arg\max_\theta \sum_{t=1}^{T} \ell_t(\theta), \qquad \ell_t = -\frac{1}{2}\left(\ln \sigma_t^2 + \frac{\varepsilon_t^2}{\sigma_t^2}\right)$$

  - T: number of days; ℓ_t: Gaussian log-density of day t without its constant; $\hat{\theta}$ is consistent if the variance equation is right

Prerequisites for Today (2/23): persistence, residual kurtosis and GARCH-t

- **Persistence** of GARCH(1,1) Bollerslev (1986): $\alpha + \beta$, the speed at which a variance shock dies out

$$\mathbb{E}_t\, \sigma^2_{t+h} - \bar{\sigma}^2 = (\alpha+\beta)^{h-1}\left(\sigma^2_{t+1} - \bar{\sigma}^2\right), \qquad \bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta}$$

  - $\bar{\sigma}^2$: the long-run (unconditional) variance; $\mathbb{E}_t$: the expectation given the data up to day t
  - half-life $\ln 0.5 / \ln(\alpha+\beta)$ days; $\alpha+\beta \to 1$: an almost integrated GARCH, shocks last for months

- **Kurtosis of the standardised residuals** $\hat{\eta}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$

$$\hat{\kappa}_\eta = \frac{\frac{1}{T}\sum_t \hat{\eta}_t^4}{\left(\frac{1}{T}\sum_t \hat{\eta}_t^2\right)^2}$$

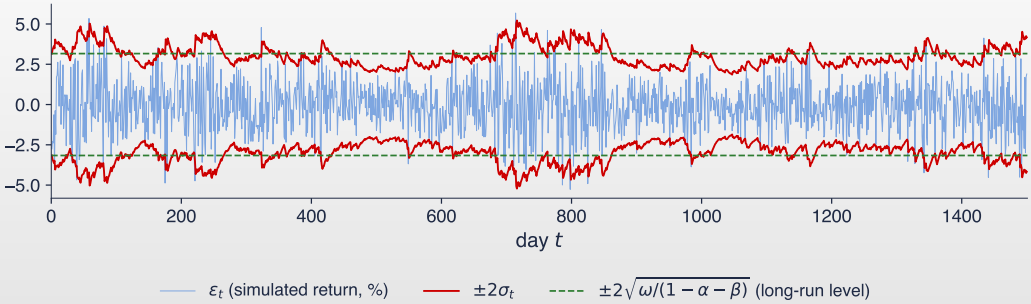
  - 3 for Normal shocks; GARCH explains part of the fat tails of returns, $\hat{\kappa}_\eta > 3$ shows the rest

- **GARCH-t**: η_t is a Student-t with $\nu > 2$ degrees of freedom, rescaled to variance 1

$$\ell_t = \ln \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)}} - \frac{1}{2}\ln\sigma_t^2 - \frac{\nu+1}{2}\ln\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right)$$

  - Γ: the gamma function; small ν: fat tails ($\kappa_\eta = 3 + 6/(\nu-4)$ for $\nu > 4$); $\nu \to \infty$: Normal
  - efficient if the t shape is right, but not robust: Gaussian QML stays consistent whatever the shape of η_t

Prerequisites for Today (3/23): a simulated GARCH(1,1) path



- Simulated: $\omega = 0.05$, $\alpha = 0.08$, $\beta = 0.90$, Normal η_t; $\alpha + \beta = 0.98$, half-life 34 days
- The band $\pm 2\sigma_t$ widens after large shocks and shrinks slowly back to the long-run level $\pm 2\bar{\sigma}$
- Volatility clustering: large returns follow large returns, of either sign

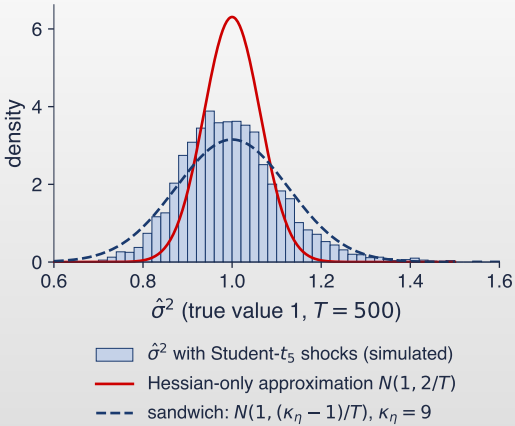
Prerequisites for Today (4/23): the sandwich

- The variance of the QML estimator is a "sandwich" of two matrices

$$\mathrm{Var}(\hat{\theta}) \approx \frac{1}{T}\, A^{-1} B\, A^{-1}, \qquad A = -\frac{1}{T}\sum_t \frac{\partial^2 \ell_t}{\partial\theta\,\partial\theta'}, \qquad B = \frac{1}{T}\sum_t s_t s_t'$$

  - $s_t = \partial \ell_t / \partial\theta$: the score of day t; A: minus the average Hessian; B: the average outer product of the scores
  - Hessian only: $\frac{1}{T}A^{-1}$; outer product of the scores (OPG) only: $\frac{1}{T}B^{-1}$; both are valid only if the likelihood is the true one ($A = B$)
- For GARCH the sandwich is $(\kappa_\eta - 1)/2$ times the Hessian-only variance
  - standard errors from the sandwich: Bollerslev–Wooldridge; with fat tails ($\kappa_\eta > 3$) the Hessian ones are too small

Prerequisites for Today (5/23): QML standard errors with fat tails



- Simplest case: $r_t = \sigma \eta_t$, $\hat{\sigma}^2 = \frac{1}{T} \sum_t r_t^2$, $T = 500$, 4 000 samples
- η_t: Student-t_5 rescaled to variance 1, so $\kappa_\eta = 9$
- The Hessian-only density (red) is far too narrow: its s.d. is $\sqrt{2/(\kappa_\eta - 1)} = 0.5$ times the right one
- The sandwich (dashed) matches the spread of the estimates

Prerequisites for Today (6/23): GARCH-MIDAS

- GARCH-MIDAS: the daily variance is a monthly level τ_t times a daily GARCH factor $g_{i,t}$ with mean 1

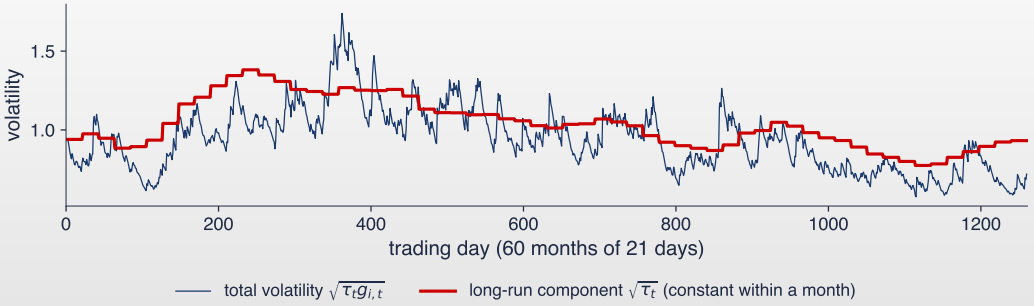
$$r_{i,t} = \mu + \sqrt{\tau_t g_{i,t}}\,\eta_{i,t}, \qquad \ln \tau_t = m + \theta \sum_{k=1}^{K} \varphi_k(w) X_{t-k}$$

  - $r_{i,t}$: return of day i of month t; μ: mean return; X_{t-k}: a monthly variable k months ago; K: number of monthly lags
  - m: intercept; θ: effect of X on the long-run variance; $\varphi_k(w) \propto (1 - k/K)^{w-1}$: decaying beta weights that sum to 1 ($w = 1$: equal weights)
- Variance ratio: the share of the variation of log volatility due to the long-run component

$$\mathrm{VR} = \frac{\mathrm{Var}(\ln \tau_t)}{\mathrm{Var}(\ln(\tau_t g_{i,t}))} \in [0, 1]$$

  - BIC $= -2 \ln L + k \ln T$, with L the maximised likelihood and k the number of parameters: lower is better

Prerequisites for Today (7/23): the two components of GARCH-MIDAS



- Simulated: X_t an AR(1) monthly variable, $K = 12$ lags, $w = 3$, $\theta = 0.6$; daily GARCH factor $\alpha = 0.07$, $\beta = 0.90$
- Red: $\sqrt{\tau_t}$, a step function that changes only at month ends, driven by the macro variable
- Blue: the daily volatility fluctuates around the red line, since $g_{i,t}$ has mean 1

Prerequisites for Today (8/23): realised measures and noise

- Realised variance from n intraday returns r_i of a day, and its precision

$$\mathrm{RV} = \sum_{i=1}^{n} r_i^2, \qquad \mathrm{Var}(\mathrm{RV} - \mathrm{IV}) \approx \frac{2\,\mathrm{IQ}}{n}, \qquad \widehat{\mathrm{IQ}} = \mathrm{RQ} = \frac{n}{3}\sum_{i=1}^{n} r_i^4$$

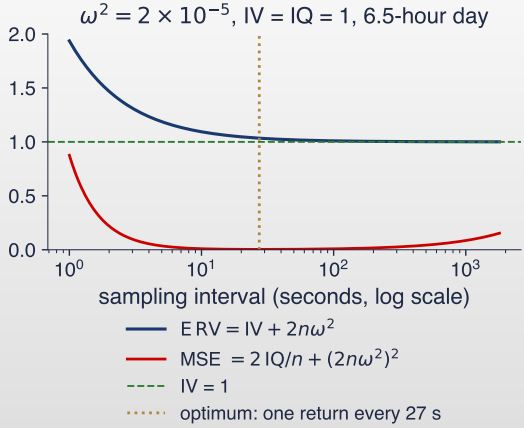
  - IV: integrated variance, the true variance of the day; without noise $\mathrm{RV} \to \mathrm{QV}$, the quadratic variation = IV plus the squared jumps
  - IQ: integrated quarticity (the daily average of σ^4); RQ: realised quarticity, its estimate

- Noise: the observed log price is $Y = X + u$

$$\mathbb{E}\,\mathrm{RV} = \mathrm{IV} + 2n\omega^2, \qquad n^* \approx \left(\frac{\mathrm{IQ}}{4\omega^4}\right)^{1/3}$$

  - X: efficient log price; u: i.i.d. noise with mean 0 and variance ω^2; the bias $2n\omega^2$ grows with the sampling frequency
  - n^*: the number of returns that minimises the approximate MSE $2\,\mathrm{IQ}/n + (2n\omega^2)^2$

Prerequisites for Today (9/23): the noise bias and the optimal sampling interval



- Constant volatility, IV = IQ = 1, i.i.d. noise $\omega^2 = 2 \times 10^{-5}$, 23 400 seconds a day
- Blue: $\mathbb{E}\,\mathrm{RV}$ rises far above IV at the finest intervals (the signature plot)
- Red: the MSE is large at both ends, bias at the left, sampling variance at the right
- Optimum $n^* = (\mathrm{IQ}/(4\omega^4))^{1/3}$ returns: here one every 27 seconds

Prerequisites for Today (10/23): two scales and the realised kernel

- Two-scales RV: the noise bias of a sparse RV is estimated from the dense RV and subtracted

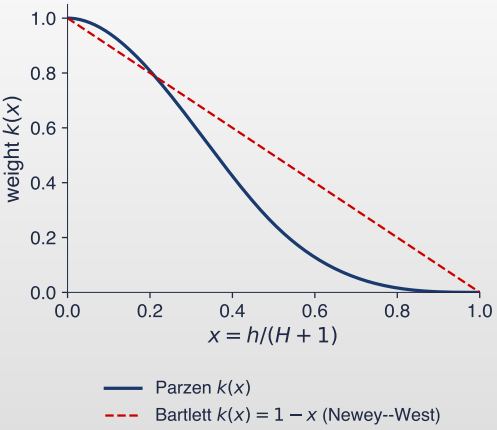
$$\mathrm{TSRV} = \overline{\mathrm{RV}}^{(K)} - \frac{\bar{n}}{n}\mathrm{RV}^{(n)}, \qquad \bar{n} = \frac{n-K+1}{K}$$

  - $\mathrm{RV}^{(n)}$: RV from all n returns; $\overline{\mathrm{RV}}^{(K)}$: the average RV over K offset sparse grids, each with about $\bar{n}$ returns
- Realised kernel: RV plus weighted autocovariances, which cancel the noise bias

$$\mathrm{RK} = \gamma_0 + 2\sum_{h=1}^{H} k\Big(\frac{h}{H+1}\Big)\gamma_h, \qquad \gamma_h = \sum_i r_i r_{i-h}$$

  - γ_h: realised autocovariance at lag h; H: bandwidth (number of lags); $k(\cdot)$: weight function
  - Parzen weights: $k(x) = 1 - 6x^2 + 6x^3$ for $x \le \frac{1}{2}$ and $2(1-x)^3$ for $\frac{1}{2} < x \le 1$

Prerequisites for Today (11/23): kernel weights



- Weight of the realised autocovariance γ_h as a function of $x = h/(H+1)$
- Parzen: flat near 0, smooth, exactly 0 at $x = 1$; it keeps the realised kernel non-negative
- Bartlett: the linear weights of the Newey–West estimator
- A larger bandwidth H removes more noise bias but adds variance

Prerequisites for Today (12/23): bipower variation and the jump test

- Bipower variation estimates IV even with jumps; tripower quarticity estimates IQ even with jumps

$$\mathrm{BV} = \mu_{\mathbf{1}}^{-\mathbf{2}} \frac{n}{n-1} \sum_{i=\mathbf{2}}^{n} |r_i||r_{i-\mathbf{1}}|, \qquad \mathrm{TQ} = n\mu_{\mathbf{4/3}}^{-\mathbf{3}} \frac{n}{n-2} \sum_{i=\mathbf{3}}^{n} \prod_{j=\mathbf{0}}^{\mathbf{2}} |r_{i-j}|^{\mathbf{4/3}}$$

  - $\mu_{\mathbf{1}} = \mathbb{E}|Z| = \sqrt{2/\pi}$, $\mu_{\mathbf{4/3}} = \mathbb{E}|Z|^{\mathbf{4/3}}$ for $Z \sim N(0,1)$: constants that make BV and TQ consistent
- Huang–Tauchen test: the relative jump contribution $(\mathrm{RV} - \mathrm{BV})/\mathrm{RV}$, studentised

$$z = \frac{(\mathrm{RV} - \mathrm{BV})/\mathrm{RV}}{\sqrt{\vartheta n^{-\mathbf{1}} \max(\mathbf{1}, \mathrm{TQ}/\mathrm{BV}^{\mathbf{2}})}} \approx N(0,1), \qquad \vartheta = \frac{\pi^{\mathbf{2}}}{4} + \pi - 5 \approx 0.609$$

  - reject "no jump on this day" if $z > z_{\mathbf{1}-\alpha}$, the $1-\alpha$ quantile of $N(0,1)$; at $\alpha = 0.1\%$, $z_{\mathbf{0.999}} = 3.09$

Prerequisites for Today (13/23): many jump tests: false rejections and the jump share

- **Expected false rejections**: one test per day, T days, level α

$$\mathbb{E}\big[\#\{t : z_t > z_{\mathbf{1}-\alpha}\}\big] = \alpha T \quad \text{if no day has a jump}$$

  - z_t: the jump statistic of day t; at $\alpha = 0.1\%$ and $T = 3\,000$: 3 days are expected by chance
- **Bonferroni**: the probability of at least one false rejection over all days is at most 5%

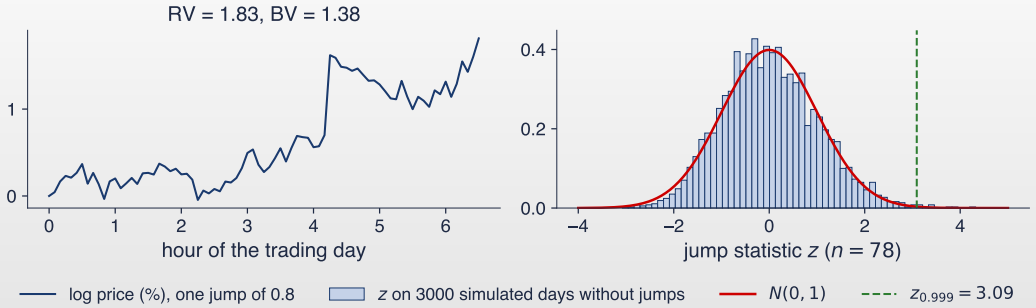
$$z_t > z_{\mathbf{1}-\mathbf{0.05}/T}$$

  - a much higher threshold: with $T = 3\,000$, $z_{\mathbf{1}-\mathbf{0.05}/T} \approx 4.15$
- **Jump share of total variation**: the RV of significant days that BV does not explain

$$\text{jump share} = \frac{\sum_t 1\{z_t > z_{\mathbf{1}-\alpha}\} \max(\mathrm{RV}_t - \mathrm{BV}_t, 0)}{\sum_t \mathrm{RV}_t}$$

  - $1\{\cdot\}$: 1 if the condition holds, else 0; many jump days with a small share: frequent but small jumps

Prerequisites for Today (14/23): a jump inside a day and the jump statistic



- Left: one simulated day, 78 five-minute returns with IV = 1 and one jump of 0.8: RV counts the jump, BV almost not
- Right: z on 3 000 simulated days without jumps is close to $N(0, 1)$; days above 3.09 are false rejections

Prerequisites for Today (15/23): HAR and HARQ

- HAR: tomorrow's RV regressed on the daily, weekly and monthly averages of past RV

$$\mathrm{RV}_{t+\mathbf{1}} = \beta_{\mathbf{0}} + \beta_d \mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + u_{t+\mathbf{1}}$$

  - $\mathrm{RV}_t^{(w)}$, $\mathrm{RV}_t^{(m)}$: means of RV over the last 5 and 22 days; $\beta_d + \beta_w + \beta_m$: persistence; $u_{t+\mathbf{1}}$: error
  - OLS with Newey–West (HAC) standard errors, robust to heteroskedasticity and autocorrelation
- Errors in variables: if $\mathrm{RV}_t = \mathrm{IV}_t + e_t$ and $\mathrm{IV}_{t+\mathbf{1}} = c + \phi \mathrm{IV}_t + v_{t+\mathbf{1}}$, the OLS slope is attenuated

$$\operatorname{plim} \hat{\phi} = \phi\lambda, \qquad \lambda = \frac{\mathrm{Var}(\mathrm{IV})}{\mathrm{Var}(\mathrm{IV}) + \mathrm{Var}(e)}$$

  - e_t: measurement error of RV; λ: reliability ratio, the share of signal in the variance of RV
  - HARQ: β_d is replaced by $\beta_d + \beta_{dQ}\sqrt{\mathrm{RQ}_t}$, so yesterday's weight falls on imprecise days ($\beta_{dQ} < 0$)

Prerequisites for Today (16/23): estimating and comparing HAR models

- **Fit**: R^2, the share of the variance of RV_{t+1} explained by the regression

$$R^2 = 1 - \frac{\sum_t \hat{u}_t^2}{\sum_t (\mathrm{RV}_{t+1} - \overline{\mathrm{RV}})^2} \in [0, 1]$$

  - $\hat{u}_t$: OLS residuals; $\overline{\mathrm{RV}}$: the sample mean of RV

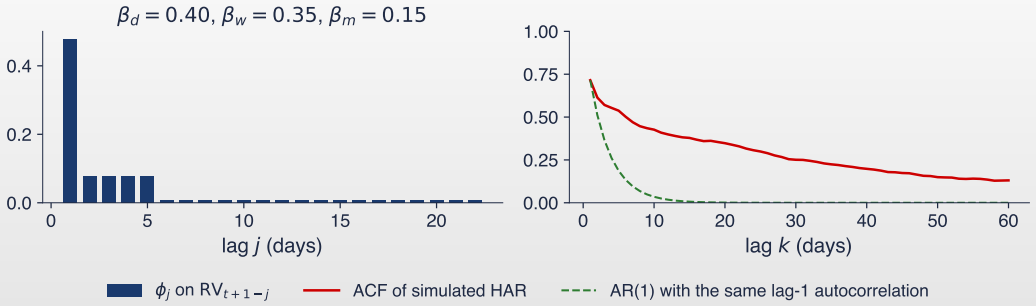
- **log-HAR**: the same regression on $\ln \mathrm{RV}$, back-transformed with a variance correction

$$\ln \mathrm{RV}_{t+1} = \beta_0 + \beta_d \ln \mathrm{RV}_t + \beta_w (\ln \mathrm{RV})_t^{(w)} + \beta_m (\ln \mathrm{RV})_t^{(m)} + u_{t+1}, \qquad \hat{h}_{t+1} = e^{\widehat{\ln \mathrm{RV}}_{t+1} + \hat{s}_u^2/2}$$

  - $(\ln \mathrm{RV})_t^{(w)}$, $(\ln \mathrm{RV})_t^{(m)}$: means of $\ln \mathrm{RV}$ over the last 5 and 22 days
  - $\hat{s}_u^2$: the residual variance; the logs tame the spikes of RV and keep the forecast positive

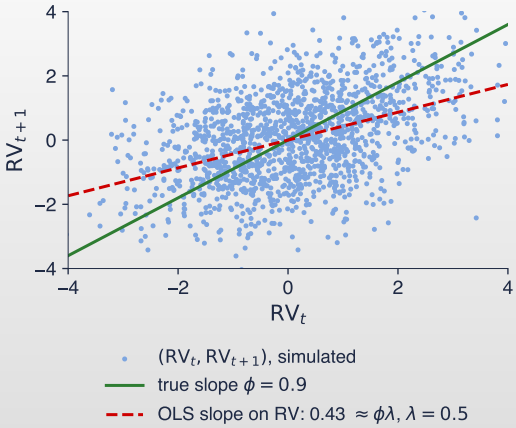
- **Benchmarks and comparison**
  - random walk: $\hat{h}_{t+1} = \mathrm{RV}_t$; rolling window: each forecast uses only the last 1 000 days
  - HAR is a restricted AR(22): $\mathrm{RV}_{t+1} = \beta_0 + \sum_{j=1}^{22} \phi_j \mathrm{RV}_{t+1-j} + u_{t+1}$ with three distinct steps in ϕ_j

Prerequisites for Today (17/23): the HAR lag structure



- Left: the AR(22) coefficients implied by $\beta_d = 0.40$, $\beta_w = 0.35$, $\beta_m = 0.15$: a large weight on yesterday, a step up to day 5, a long flat tail
- Right: the ACF of a simulated HAR decays slowly, like long memory, while an AR(1) with the same first autocorrelation dies out in two weeks

Prerequisites for Today (18/23): measurement error and the attenuation of the slope



- Simulated: IV_t an AR(1) with $\phi = 0.9$ and variance 1; $\mathrm{RV}_t = \mathrm{IV}_t + e_t$, $\mathrm{Var}(e_t) = 1$
- $\lambda = 1/(1+1) = 0.5$: the OLS slope on RV is about $\phi\lambda = 0.45$, not 0.9
- Noisy days get too much weight; HARQ lowers the weight of yesterday when RQ is high

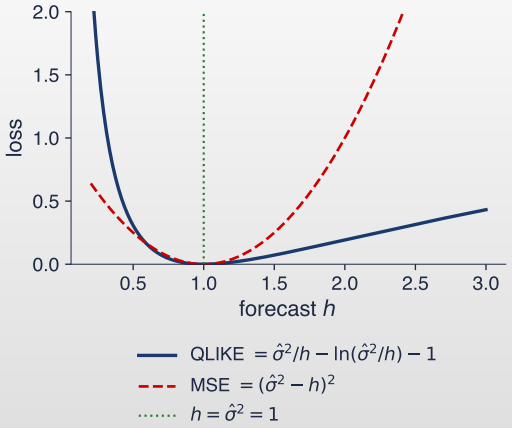
Prerequisites for Today (19/23): evaluating volatility forecasts

- Two robust losses for a forecast h against a proxy $\hat{\sigma}^2$ (Patton 2011)

$$\mathrm{QLIKE} = \frac{\hat{\sigma}^2}{h} - \ln\frac{\hat{\sigma}^2}{h} - 1, \qquad \mathrm{MSE} = (\hat{\sigma}^2 - h)^2$$

  - proxy: an unbiased but noisy measure of the true variance (squared return, RV); robust: the ranking of forecasts is the one the true variance would give
- For $z \sim N(0,1)$: $\mathbb{E}\ln z^2 = \psi(\frac{1}{2}) + \ln 2 \approx -1.27$ and $\mathrm{Var}\ln z^2 = \pi^2/2$
  - ψ: the digamma function, the derivative of $\ln\Gamma$; these moments are needed for the MSE-log loss $(\ln\hat{\sigma}^2 - \ln h)^2$
- Insanity filter: a forecast outside the in-sample range of RV is replaced by the in-sample mean
- Diebold–Mariano (DM): t-statistic of the mean loss differential with a HAC variance; $|DM| > 1.96$: significant difference at 5%

Prerequisites for Today (20/23): the shape of QLIKE and MSE



- Both losses for a proxy $\hat{\sigma}^2 = 1$, as functions of the forecast h; both are 0 at $h = \hat{\sigma}^2$
- QLIKE punishes under-prediction much more than over-prediction
- MSE is symmetric and depends on the level: days with high variance dominate the average
- QLIKE depends only on the ratio $\hat{\sigma}^2/h$: calm and turbulent days count alike

Prerequisites for Today (21/23): DCC and cDCC

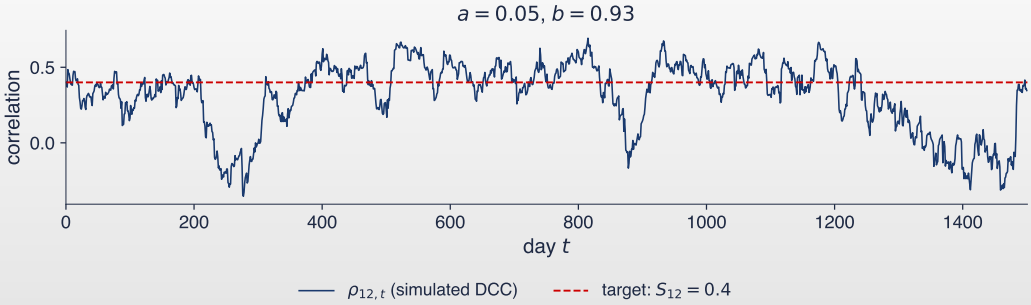
- DCC: a GARCH-type recursion for a quasi-correlation matrix Q_t of the standardised residuals

$$Q_t = (1 - a - b)S + a\, z_{t-\mathbf{1}} z_{t-\mathbf{1}}' + b\, Q_{t-\mathbf{1}}, \qquad \rho_{\mathbf{12},t} = \frac{q_{\mathbf{12},t}}{\sqrt{q_{\mathbf{11},t}\, q_{\mathbf{22},t}}}$$

  - z_t: vector of returns divided by their own GARCH volatilities; S: sample correlation matrix of z_t (the target)
  - a: reaction to yesterday's co-movement; b: persistence; $q_{ij,t}$: elements of Q_t; $\rho_{\mathbf{12},t}$: the conditional correlation of assets 1 and 2

- cDCC: $z_{t-\mathbf{1}}$ is replaced by $z^*_{t-\mathbf{1}} = \mathrm{diag}(Q_{t-\mathbf{1}})^{\mathbf{1/2}} z_{t-\mathbf{1}}$
  - $\mathrm{diag}(Q_{t-\mathbf{1}})^{\mathbf{1/2}}$: diagonal matrix with $\sqrt{q_{ii,t-\mathbf{1}}}$; the correction makes the estimator of S consistent

Prerequisites for Today (22/23): a simulated DCC correlation



- Simulated DCC: $a = 0.05$, $b = 0.93$, target correlation $S_{12} = 0.4$; $a + b = 0.98$: correlation shocks last for weeks
- $\rho_{12,t}$ wanders far from 0.4 and returns to it: a constant correlation would hide these swings

Prerequisites for Today (23/23): Realized GARCH and annualised realised volatility

- **Realized GARCH** Hansen, Huang and Shek (2012): the realised measure x_t enters the variance equation and has its own measurement equation

$$\ln h_t = \omega + \beta \ln h_{t-\mathbf{1}} + \gamma \ln x_{t-\mathbf{1}}, \qquad \ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$$

  - h_t: the conditional variance of the return $r_t = \sqrt{h_t}\, z_t$; x_t: e.g. RV or the realised kernel
  - ξ, φ: bias of x_t ($\xi = 0$, $\varphi = 1$: unbiased); u_t: measurement noise; $\tau(z) = \tau_{\mathbf{1}} z + \tau_{\mathbf{2}}(z^{\mathbf{2}} - 1)$: leverage

- **Annualised volatility** from a daily realised measure

$$\sigma_{\textbf{annual}} = \sqrt{A \cdot \frac{1}{T} \sum_{t=\mathbf{1}}^{T} \mathrm{RK}_t}$$

  - A: periods per year, 252 for exchanges, 365 for crypto-assets (traded every day); RK: the realised kernel

A1: the QML variance and its robust standard error [Solved]

Task

- BET daily returns (%), demeaned, 2016–2026, $T = 2\,674$; model $r_t = \sigma \eta_t$ with η_t i.i.d., mean 0, variance 1, kurtosis κ.
- 1. Derive the QMLE $\hat{\sigma}^2$ and its Hessian-only variance under normality.
- 2. Derive the sandwich variance and compute both standard errors from $\hat{\sigma}^2 = 0.993$ and $\hat{\kappa} = 20.79$.
- 3. State by how much the Hessian understates the uncertainty.
- Report: one estimator, two standard errors, one ratio, one sentence.

Solution

- 1. $\ell_t = -\frac{1}{2}(\ln \sigma^2 + r_t^2/\sigma^2)$; $\hat{\sigma}^2 = \frac{1}{T}\sum r_t^2$; $A = \frac{1}{2\sigma^4}$, so the Hessian-only variance is $2\sigma^4/T$
- 2. $s_t = \frac{r_t^2 - \sigma^2}{2\sigma^4}$, $B = \frac{(\kappa-1)\sigma^4}{4\sigma^8}$; $A^{-1} B A^{-1} = (\kappa - 1)\sigma^4$: s.e. 0.0272 (Hessian) and 0.0854 (sandwich)
- 3. Ratio $\sqrt{(\hat{\kappa} - 1)/2} = 3.15$: with the BET's fat tails, Hessian intervals are about three times too narrow

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A2: a sandwich by hand [Proposed]

Solution discussed in class

Task

- GARCH(1,1) by QML, $T = 4000$; per-observation averages for (α, β): $A = \begin{pmatrix} 50 & 40 \\ 40 & 36 \end{pmatrix}$, $B = \begin{pmatrix} 120 & 90 \\ 90 & 72 \end{pmatrix}$. Model: A1.
- 1. Compute A^{-1} and the Hessian-only standard errors $\sqrt{\mathrm{diag}(A^{-1})/T}$.
- 2. Compute $A^{-1} B A^{-1}$ and the sandwich standard errors.
- 3. Compute the OPG standard errors and say which of the three you would report.
- Report: three pairs of standard errors, one sentence.

A3: the noise bias and the optimal sampling frequency [Solved]

Task

- A 6.5-hour day, constant volatility with IV = IQ = 1 ($\%^2$, $\%^4$), i.i.d. noise with $\omega^2 = 1.6 \times 10^{-5}$.
- 1. Compute $\mathbb{E}\,\mathrm{RV}$ at 1 second ($n = 23\,400$), 1 minute ($n = 390$) and 5 minutes ($n = 78$).
- 2. Compute the approximate MSE of the primer slide Realised measures and noise at the three frequencies.
- 3. Derive n^* and convert it into seconds.
- Report: three expectations, three MSE values, one optimal interval.

Solution

- 1. $\mathbb{E}\,\mathrm{RV} = 1 + 2n\omega^2$: 1.7488, 1.0125, 1.0025
- 2. MSE: 0.5608 (bias dominates), 0.0053, 0.0256 (variance dominates)
- 3. $\frac{d}{dn}\left[\frac{2}{n} + 4n^2\omega^4\right] = 0 \Rightarrow n^* = (4\omega^4)^{-1/3} = 992$: one return every 23.6 seconds

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A4: two scales and kernel weights [Proposed]

Solution discussed in class

Task

- One day: $\mathrm{RV}^{(n)} = 1.75$ at one second ($n = 23\,400$), average five-minute RV over the $K = 300$ offset grids $\overline{\mathrm{RV}}^{(K)} = 1.005$. Model: A3.
- 1. Compute $\bar{n}$, TSRV and its small-sample version $\mathrm{TSRV}/(1 - \bar{n}/n)$.
- 2. Estimate ω^2 by $\mathrm{RV}^{(n)}/(2n)$ and say why it is biased upwards.
- 3. Compute the Parzen weights $k(h/5)$ for $h = 1, \ldots, 4$ ($H = 4$).
- Report: three numbers, one estimate and a sentence, four weights.

A5: RV, BV and the jump test on a toy day [Solved]

Task

- Eight returns (%): $0.10, -0.12, 0.08, 1.20, -0.09, 0.11, -0.07, 0.10$.
- 1. Compute RV and BV; compute RV without the fourth return.
- 2. Compute TQ, the ratio $(\mathrm{RV} - \mathrm{BV})/\mathrm{RV}$ and the Huang–Tauchen z.
- 3. Decide at 0.1% and explain why $n = 8$ makes the test weak.
- Report: three numbers, three numbers, one decision and one sentence.

Solution

- 1. RV $= 1.5059$; BV $= 0.4492$; without the jump RV $= 0.0659$: the jump dominates RV, BV is far less affected
- 2. TQ $= 0.1360$, $\mathrm{TQ}/\mathrm{BV}^2 < 1$ so max = 1; ratio 0.7017; $z = 0.7017/\sqrt{0.609/8} = 2.54$
- 3. $z < 3.09$: no rejection at 0.1% ($p = 0.0055$); with eight returns BV still carries the jump through two products, and ϑ/n is large

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A6: the smallest detectable jump [Proposed]

Solution discussed in class

Task

- Constant volatility, IV $= 1$, so $\mathrm{BV} \approx 1$, $\mathrm{TQ}/\mathrm{BV}^2 \approx 1$; one jump of size J (in units of $\sqrt{\mathrm{IV}}$), so $\mathrm{RV} \approx 1 + J^2$. Model: A5.
- 1. Write z as a function of J and n.
- 2. Solve $z = 3.09$ for J with $n = 78$ and $n = 390$.
- 3. Compare with the power curve of the lecture and say what limits finer sampling.
- Report: one formula, two thresholds, one sentence.

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A7: HAR as a restricted AR(22) [Solved]

Task

- HAR with $\beta_{\mathbf{0}} = 0.10$, $\beta_d = 0.30$, $\beta_w = 0.40$, $\beta_m = 0.20$.
- 1. Write the implied AR(22) coefficients ϕ_j on $\mathrm{RV}_{t+\mathbf{1}-j}$.
- 2. Compute their sum and the unconditional mean of RV.
- 3. Count the restrictions HAR imposes on the AR(22).
- Report: three distinct coefficients, two numbers, one count.

Solution

- 1. $\phi_{\mathbf{1}} = \beta_d + \frac{\beta_w}{\mathbf{5}} + \frac{\beta_m}{\mathbf{22}} = 0.3891$;
  $\phi_{\mathbf{2..5}} = \frac{\beta_w}{\mathbf{5}} + \frac{\beta_m}{\mathbf{22}} = 0.0891$; $\phi_{\mathbf{6..22}} = \frac{\beta_m}{\mathbf{22}} = 0.0091$
- 2. $\sum \phi_j = \beta_d + \beta_w + \beta_m = 0.90$; mean $\beta_{\mathbf{0}}/(1 - \sum \phi_j) = 1.00$
- 3. 23 coefficients reduced to 4: 19 linear restrictions (three steps)

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A8: attenuation and HARQ [Proposed]

Solution discussed in class

Task

- $\mathrm{IV}_{t+\mathbf{1}} = c + 0.95\,\mathrm{IV}_t + v_{t+\mathbf{1}}$, $\mathrm{Var}(\mathrm{IV}) = 0.8$; $\mathrm{RV}_t = \mathrm{IV}_t + e_t$ with $\mathbb{E}\mathrm{Var}(e_t) = 0.2$; $n = 78$. Model: A7.
- 1. Compute λ and the probability limit of the OLS slope of $\mathrm{RV}_{t+\mathbf{1}}$ on RV_t.
- 2. Compute λ_t and the slope on a quiet day ($\mathrm{IQ}_t = 0.5$) and on a turbulent day ($\mathrm{IQ}_t = 5$), with $\mathrm{Var}(e_t) = 2\mathrm{IQ}_t/n$.
- 3. Explain the sign of β_{dQ} in HARQ.
- Report: two numbers, four numbers, one sentence.

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A9: robust and non-robust losses against the squared return [Solved]

Task

- $r_t = \sigma_t z_t$, $z_t \sim N(0,1)$, $\sigma_t^2 = 1$; proxy $\hat{\sigma}_t^2 = r_t^2$; forecasts $h = 1$ (true) and $h = 0.6$ (biased).
- 1. Find the forecast that minimises $\mathbb{E}(\ln r_t^2 - \ln h)^2$.
- 2. Compute the expected MSE-log and QLIKE losses of both forecasts.
- 3. Say which loss ranks the forecasts correctly.
- Report: one number, four numbers, one sentence.

Solution

- 1. $h^* = \exp(\mathbb{E} \ln z^2) = \exp(-1.2704) = 0.2807$: far below the true variance
- 2. MSE-log: $\frac{\pi^2}{2} + (\mathbb{E} \ln z^2 - \ln h)^2$: 6.5486 ($h = 1$) and 5.5117 ($h = 0.6$); QLIKE: $\frac{1}{h} + 1.27 + \ln h - 1$: 1.2704 and 1.4262
- 3. MSE-log prefers the biased forecast; QLIKE (and MSE: 2 against 2.16) prefer the true one

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A10: one DCC and one cDCC update [Proposed]

Solution discussed in class

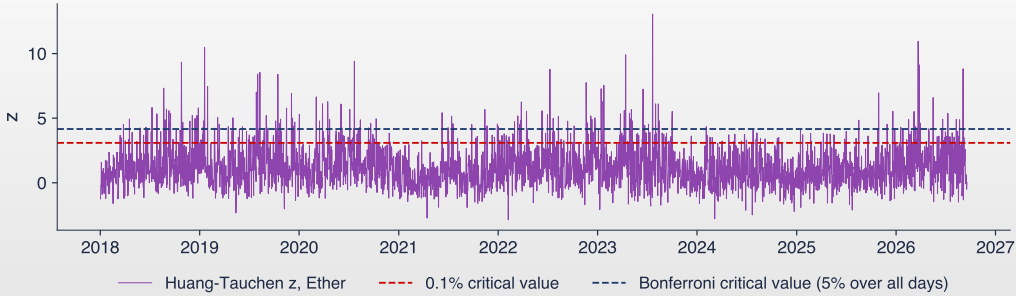
Task

- $S = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$, $a = 0.05$, $b = 0.93$, $Q_{t-\mathbf{1}} = \begin{pmatrix} 1.2 & 0.7 \\ 0.7 & 0.9 \end{pmatrix}$, $z_{t-\mathbf{1}} = (1.5, -0.5)'$. Model: A9.
- 1. Compute Q_t and $\rho_{\mathbf{12},t}$ with DCC.
- 2. Compute $z^*_{t-\mathbf{1}}$, Q_t and $\rho_{\mathbf{12},t}$ with cDCC.
- 3. Compare with $\rho_{\mathbf{12},t-\mathbf{1}}$ and explain the direction of the change.
- Report: a matrix and a correlation, a vector, a matrix and a correlation, one sentence.

B1: jumps in Ether [Solved]

- **Question**: how often does Ether jump, and how much of its variation do jumps explain?
- **Inputs**: our daily RV, BV and TQ from 5-minute returns ($n = 288$ a UTC day), 2018–2026, $T = 3\,165$
- **Tasks**
  1. Compute the Huang–Tauchen statistic for every day and count the days above the 0.1% critical value.
  2. Compare with the number of false rejections expected under the null, and repeat with a Bonferroni threshold for 5% over all days.
  3. Compute the share of total RV due to the significant jumps.
  4. Interpretation: are jumps frequent, large, or both?
- **Report**: two counts, one expected number, one share, one sentence
- Notebook: section B1

B1: solution [Solved]



- At 0.1%: 355 jump days (11.2%) where 3.2 false rejections are expected; Bonferroni threshold $z > 4.16$: still 137 days
- Largest statistics: 22.07.2023 ($z = 13.1$), 23.03.2026 (10.9), 19.01.2019 (10.5); jump share of total RV 2.3%
- Interpretation: rejections are far too frequent for the null (mean $z = 1.19$), but jumps carry a small share of variation: frequent small "jumps" partly reflect intraday seasonality and zero returns, a few days are genuine large jumps

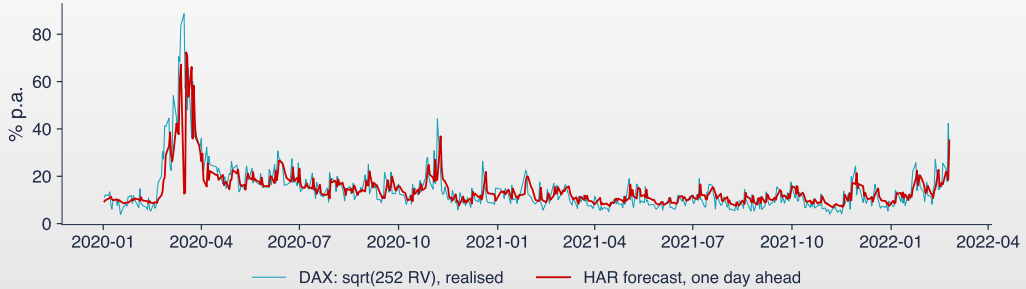
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B2: HAR for the DAX [Solved]

- **Question**: how persistent is DAX realised variance, and does HAR beat the random walk out of sample?
- **Inputs**: Oxford-Man 5-minute RV of the DAX ($\%^2$), 2000–2022, $T = 5\,589$; rolling window of 1 000 days with the insanity filter
- **Tasks**
  1. Estimate HAR by OLS on the full sample and report Newey–West standard errors (5 lags), R^2 and the persistence.
  2. Produce one-day-ahead HAR, log-HAR and random-walk (RV_t) forecasts and compute the average QLIKE.
  3. Test HAR against each alternative with Diebold–Mariano.
  4. Interpretation: which horizon component matters most for the DAX?
- **Report**: four coefficients with s.e., two statistics, three average losses, two DM statistics, one sentence
- Notebook: section B2

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B2: solution [Solved]



- $\hat{\beta}_d = 0.414$ (0.069), $\hat{\beta}_w = 0.279$ (0.107), $\hat{\beta}_m = 0.210$ (0.060), $\hat{\beta}_0 = 0.151$; $R^2 = 0.561$; persistence 0.903
- Out of sample from 22.01.2004 (4 589 days): QLIKE HAR 0.1863, random walk 0.2280 (DM 2.75), log-HAR 0.1597 (DM −2.09)
- Interpretation: the daily component carries the largest weight for the DAX, the monthly one keeps the forecast anchored; HAR beats the random walk clearly and is beaten by its log version

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B3: HARQ for Ether out of sample [Proposed]

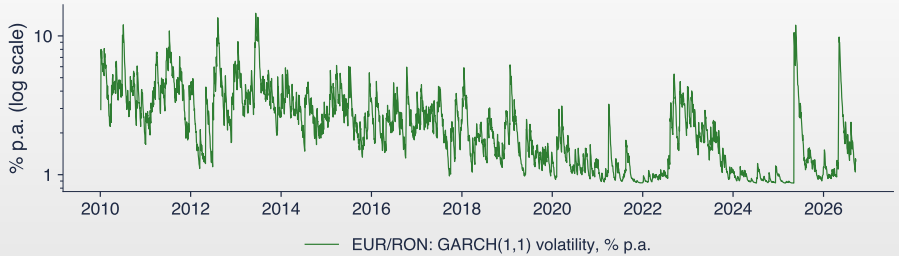
- **Question**: does the HARQ gain of Bollerslev, Patton and Quaedvlieg (2016) hold for Ether?
- **Inputs**: Ether 5-minute RV and RQ, 2018–2026; rolling window of 1 000 days; model: B2 and A8
- **Tasks**
  1. Produce HAR and HARQ forecasts with the insanity filter and compute QLIKE and MSE ratios HARQ/HAR with DM statistics.
  2. Count the raw HARQ forecasts outside the in-sample range and the negative ones.
  3. Replace the filter by a floor at the smallest in-sample RV and repeat.
  4. Interpretation: is the gain robust to the treatment of implausible forecasts?
- **Report**: four ratios with DM statistics, two counts, one sentence
- Notebook: section B3

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B4: QML for EUR/RON [Solved]

- **Question**: how persistent is EUR/RON volatility, and how much does the choice of standard errors matter?
- **Inputs**: daily log returns of the BNR reference rate (%), demeaned, 2010–2026, $T = 4\,207$
- **Tasks**
  1. Estimate GARCH(1,1) by Gaussian QML; report Hessian and Bollerslev–Wooldridge standard errors.
  2. Compute the kurtosis of the standardised residuals and compare the ratio of standard errors with $\sqrt{(\hat{\kappa} - 1)/2}$.
  3. Estimate GARCH-t by ML and compare the log-likelihoods.
  4. Interpretation: what does the persistence say about a managed float?
- **Report**: a table of three parameters with two s.e. each, one kurtosis, one log-likelihood gain, one sentence
- Notebook: section B4

B4: solution [Solved]



- $\hat{\omega} = 0.00049$ (0.00004; 0.00027), $\hat{\alpha} = 0.166$ (0.015; 0.040), $\hat{\beta} = 0.833$ (0.012; 0.027); in brackets: Hessian; sandwich
- Kurtosis of standardised residuals 61.8 (raw returns 15.4); sandwich/Hessian ratio for α 2.71; GARCH-t: $\hat{\nu} = 7.9$, log-likelihood higher by 861.0
- Interpretation: persistence 0.9998, almost integrated: long calm periods with rare bursts (2012–2013, late 2022, 2025, 2026); the extreme kurtosis even after GARCH filtering shows that one variance equation cannot describe a managed exchange rate

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B5: GARCH-MIDAS for the BET [Proposed]

- **Question**: does Romanian industrial production drive the long-run volatility of the BET?
- **Inputs**: BET daily returns from 2003; Eurostat industrial production (B–D, seasonally and calendar adjusted), monthly log growth; $K = 24$ lags; model: B4
- **Tasks**
  1. Estimate GARCH-MIDAS with log τ_t and restricted beta weights by QML; report $\hat{\theta}$ and $\hat{w}$ with sandwich standard errors.
  2. Compute the variance ratio.
  3. Compare with GARCH(1,1) on the same days by BIC.
  4. Interpretation: is the macro channel visible in Romanian stock volatility?
- **Report**: two estimates with s.e., one ratio, two BIC values, one sentence
- Notebook: section B5

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C1: Bitcoin before and after the spot ETFs [Proposed]

- **Question**: did Bitcoin volatility become more "equity-like" after the US spot ETFs started trading on 11 January 2024?
- **Inputs**: our Bitcoin realised measures, 2021–2026; S&P 500 daily returns; model: B1–B3
- **Tasks**
  1. Write a five-line pre-registration: hypothesis, sample split, statistics, tests, what would falsify the claim.
  2. For 2021–2023 and from 11 January 2024, compute annualised volatility (from the realised kernel), the share of jump days, the jump share of RV, the HAR persistence and the correlation with the S&P 500.
  3. Say which differences you could attribute to the ETFs and which to other events.
  4. Interpretation: what can and cannot be concluded, and how would you turn this into a project?
- **Report**: a five-line plan, a table of two rows, a project plan
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise volatility modelling with high-frequency data. The answer:
- (a) Gaussian QML is inconsistent when returns are fat-tailed; use Student-t ML instead.
- (b) Sampling prices as often as possible always makes realised variance more precise.
- (c) On a day without jumps, bipower variation equals realised variance.
- (d) HAR is a long-memory model.
- (e) MSE on log variances is a robust loss for comparing forecasts against squared returns.
- (f) In DCC the target S is the unconditional correlation of the standardised residuals, so the estimator is consistent.
- (g) In Realized GARCH the realised measure may be a biased and noisy signal of the conditional variance.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, a number from today's tasks (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- QML needs only the variance equation; its standard errors need the kurtosis: use the sandwich
- Realised variance trades variance against noise bias; kernels and two scales use all the data
- Jump tests reject often in real data; jumps explain little of the variation
- HAR is a restricted AR(22); HARQ lets the precision of RV set yesterday's weight
- Rank volatility forecasts with QLIKE or MSE; an AI answer is a draft to audit

After the seminar

- Lecture 8 develops each topic of today
  - QML theory, component GARCH and GARCH-MIDAS, the CLT of RV, noise and kernels, jump tests
  - HAR and HARQ, Realized GARCH, robust losses, DCC, cDCC and DCC-NL
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: the realised volatility of crypto-assets around a market-structure change
- Reading: Bollerslev and Wooldridge (1992); Barndorff-Nielsen and Shephard (2002); Corsi (2009); Bollerslev, Patton and Quaedvlieg (2016); Patton (2011)
- **The seminar is for practice and is not graded**

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